When Fairness Metrics Disagree Under Shift: A Stylized Case Study of a Medical Follow-up Risk Score under Hospital Deployment Shift

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1 Introduction

Group fairness metrics are typically defined under a fixed data-generating process, yet real-world deployments—such as moving a medical follow-up risk score across hospitals—involve distribution shifts that can make these metrics behave unpredictably. In this project, we ask: (1) how do standard group fairness metrics behave under controlled shifts even when the underlying classifier remains fixed? (2) when demographic parity and error-based metrics disagree, does the discrepancy reflect degradation with respect to the clean synthetic outcome Y^*, or measurement distortion induced by the recorded outcome $\tilde{Y}$? and (3) how do importance reweighting, post-hoc threshold adjustment, and target-distribution retraining affect fairness distortion and what trade-offs do they incur? We address these questions via a synthetic simulation framework mimicking a cross-hospital deployment.

2 Related Work

Fairness under non-stationary environments has been studied in several recent works. Asiaee and Aryan (2026) examined invariance under group-conditional prior shifts, while Chen et al. (2022) studied fairness transferability under bounded shifts. Barrainkua et al. (2022) and Shao et al. (2024) surveyed fairness under distribution shifts. Existing work rarely separates clean-outcome degradation from label-measurement distortion when selection-rate and error-based metrics disagree, which is the central gap addressed here.

3 Method

3.1 Distribution shift mechanisms

We study group-conditioned base-rate shift, pure group-specific covariate shift, and asymmetric label noise. The fixed source classifier is logistic regression on $X = (X_1, X_2)$ without group S; X_1 is informative and X_2 is noise. Equal source base rates make this model correctly specified, and pure covariate shift preserves its label conditional. Label-noise experiments retain both Y^* and the recorded outcome $\tilde{Y}$. Full constructions are given in Appendix A.

3.2 Fairness and calibration metrics

We report demographic parity difference (DP), equalized odds difference (EO), the true-positive-rate gap, balanced accuracy, and global/group-wise expected calibration error (ECE). Formal definitions are provided in Appendix B.

3.3 Mitigation strategies under different target-data access regimes

The interventions represent distinct target-data regimes rather than interchangeable fixes: KDE reweighting uses unlabeled target covariates and is justified only under pure covariate shift; equal-opportunity thresholding uses a labeled target calibration set $(X, \tilde{Y}, S)$; and target ERM uses labeled target training data $(X, \tilde{Y})$, with S needed only for fairness evaluation. Appendix C gives full details.

4 Experimental Setup

4.1 Shift parameter grid

We use 11 values of $\alpha \in [0, 1]$, 10 values of $\gamma \in [1, 4]$, and 10 values of $\beta \in [0, 0.3]$. One-dimensional results are means over three coupled seeds; the primary 11×10 (α, β) danger-zone grid uses five seeds. Appendix D reports implementation details.

5 Results

5.1 Key empirical findings

Clean-outcome degradation (RQ1–RQ2). Averaged over three seeds, base-rate shift raises DP from 0.014 to 0.252 while the TPR gap remains between 0.027 and 0.043. Under pure covariate shift, DP reaches 0.139 and EO/TPR gap reach 0.219, while ECE gap remains below 0.009.

Measurement distortion (RQ2). Under asymmetric label noise, DP remains 0.014 and predictions and Y^* do not change, yet the recorded-label TPR gap ends at 0.150 versus 0.034 without noise. The apparent degradation therefore arises from evaluating against $\tilde{Y}$ rather than a change in clean-outcome performance.

Mitigation (RQ3). Under matched pure covariate shift, KDE nearly coincides with the correctly specified frozen model, as expected when reweighting preserves the same population optimum. Equal-opportunity thresholding lowers maximum TPR gap from 0.219 to 0.044 but raises maximum EO to 0.333; at $\gamma = 4$, balanced accuracy falls from 0.893 to 0.863. Target ERM also remains close to the frozen model. Figure 2 in Appendix E shows three-seed variability; KDE under other shifts is a negative control.

5.2 Operational definition of the danger zone and phase-diagram analysis

Relative to the no-shift cell, define

$$\Delta_D = \mathrm{DP}(\alpha, \beta) - \mathrm{DP}(0, 0), \qquad \Delta_T = \mathrm{TPRGap}_{\tilde{Y}}(\alpha, \beta) - \mathrm{TPRGap}_{\tilde{Y}}(0, 0).$$

The operational *metric-monitoring danger zone* is

$$\Delta_T \geq 0.05 \quad \wedge \quad [\Delta_D \leq 0.02 \ \vee \ \Delta_T \geq 2 \max(\Delta_D, 0)] \, .$$

The cutoffs operationalize five-point TPR-gap harm, no greater than two-point DP worsening, or twofold relative deterioration; they are not clinical or significance standards (Appendix D). A cell is marked when at least three of five seeds satisfy the rule. Figure 1 shows 20 dangerous cells out of 110, emerging from

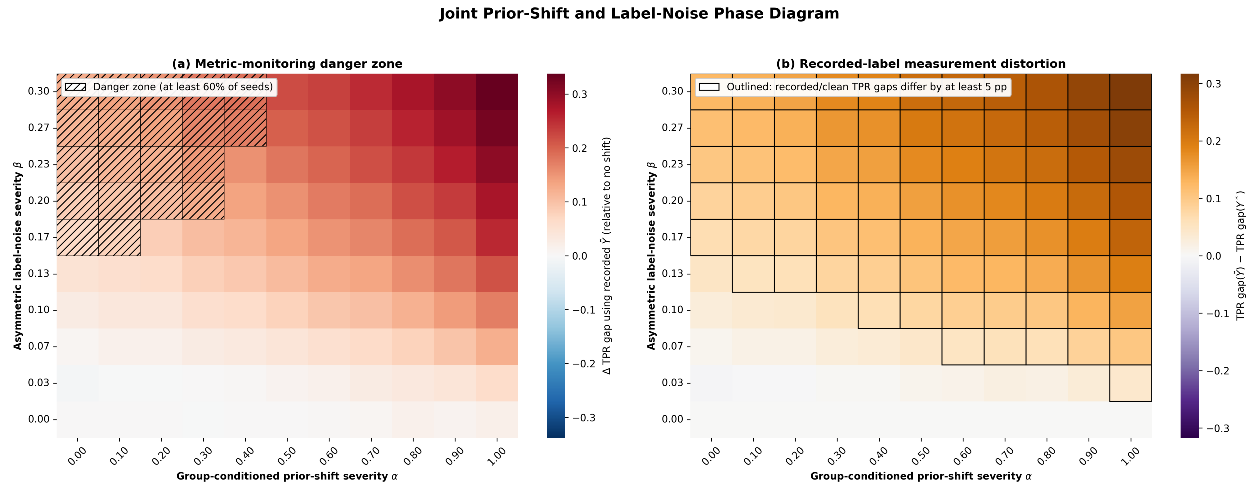

Figure 1: Primary joint prior-shift and label-noise phase diagram. Panel (a) shows the change in the recorded-label TPR gap and hatches cells satisfying the danger-zone definition in at least three of five seeds. Panel (b) shows the difference between recorded-label and clean-outcome TPR gaps; outlined cells differ by at least five percentage points in at least three seeds.

$\beta \approx 0.17$ at lower α. Recorded and clean TPR gaps differ by at least five percentage points in 78 cells; their maximum increases are 0.338 and 0.021, respectively.

Interpretation. The danger zone first appears at $\beta = 0.167$ for $\alpha \leq 0.1$ and expands to $\alpha \leq 0.4$ as recording noise increases. Here DP does not materially worsen and may appear to improve, while the recorded-label TPR gap worsens. At larger α, DP changes make the directional rule harder to satisfy, but the clean–recorded discrepancy still spans 78 cells, far beyond the 20-cell danger zone. The panels therefore distinguish a reassuring selection-rate metric from an error-rate conclusion that depends on the outcome definition.

6 Conclusion and Future Work

Stable DP can mask recorded-label error disparities, yet treating recorded outcomes as ground truth can also mistake documentation distortion for latent clinical harm. Deployment monitoring should therefore track DP, TPR gaps, calibration, and data quality with each metric tied to its outcome definition. Mitigations must likewise be interpreted by their target-data requirements and valid shift assumptions, not as interchangeable fixes. Future work should extend the danger-zone analysis to richer feature spaces and causal outcome definitions.

References

[1] Asiaee, A., & Aryan, K. (2026). Fairness under group-conditional prior probability shift: Invariance, drift, and target-aware post-processing. arXiv preprint arXiv:2602.05144.

[2] Chen, Y., Raab, R., Wang, J., & Liu, Y. (2022). Fairness transferability subject to bounded distribution shift. arXiv preprint arXiv:2206.00129.

[3] Barrainkua, A., Gordaliza, P., Lozano, J. A., & Quadrianto, N. (2022). A survey on preserving fairness guarantees in changing environments. arXiv preprint arXiv:2211.07530.

[4] Shao, M., Li, D., Zhao, C., Wu, X., Lin, Y., & Tian, Q. (2024). Supervised algorithmic fairness in distribution shifts: A survey. arXiv preprint arXiv:2402.01327.

A Full Distribution Shift Mechanisms

This appendix provides the complete mathematical definitions for the shift families summarized in Section 3.1. Each is implemented using the base data generator `generate_data` (`src/shifts.py`).

A.1 Source model and inputs

The classifier receives $X = (X_1, X_2)$ and does not receive the group label S. In the source distribution, $X_1 \mid Y^* = y \sim \mathcal{N}(2y - 1, 1)$ is informative, whereas $X_2 \sim \mathcal{N}(0, 1)$ is independent noise. Both groups have $P(Y^* = 1 \mid S) = 0.3$, so Bayes' rule gives

$$\operatorname{logit} P(Y^* = 1 \mid X, S) = \operatorname{logit}(0.3) + 2X_1.$$

The fitted logistic regression includes an intercept and both features, so its conditional model family is correctly specified; the coefficient of X_2 is zero in the population. Group labels are used only for fairness evaluation and group-specific threshold post-processing. Because the pure covariate shift below changes $P(X, S)$ while preserving this conditional, density-ratio reweighting is matched to the shift assumption but need not improve the population classifier.

A.2 Group-conditioned base-rate shift

The function `group_shift(severity)` varies the group-conditioned base rates of the clean positive class while keeping the feature generator otherwise fixed. Concretely, we define

$$\Pr(Y^* = 1 \mid S = 0) = 0.3 + 0.2 \cdot \alpha,$$
$$\Pr(Y^* = 1 \mid S = 1) = 0.3 - 0.2 \cdot \alpha,$$

with $\alpha \in [0, 1]$. Increasing the severity therefore simultaneously increases the clean positive rate for group A and decreases it for group B. The function `group_shift` passes the corresponding base rates to `generate_data` and returns the resulting features X, labels Y^*, and group labels S.

This family of shifts induces asymmetric changes in label prevalence across groups while keeping the conditional feature generator $P(X \mid Y, S)$ fixed. Because X depends on Y, its marginal distribution can still change as the base rates change, even though the conditional generator is fixed; from the perspective of an observer who does not see Y, this may resemble a covariate shift. In a hospital deployment context, this corresponds to a scenario where the target hospital serves a patient population with a different group-specific prevalence of the underlying condition (e.g., different referral patterns or catchment demographics), while the clinical manifestation of the condition given group membership remains the same.

A.3 Group-specific covariate shift

The function `covariate_shift(severity, group)` implements a pure covariate shift that affects the scale of the informative feature for one group only. Under the source Gaussian mixture, with group-specific positive-class prior π_s, the fixed label conditional is

$$P(Y = 1 \mid X, S = s) = \sigma(\operatorname{logit}(\pi_s) + 2X_1).$$

We first draw X_1^{src} from the source marginal and, for the target group, transform it as

$$X_1^{\text{target}} = \mu_s + \gamma(X_1^{\text{src}} - \mu_s), \qquad \mu_s = 2\pi_s - 1,$$

with $\gamma \in [1, 4]$. We then sample the target label from the same conditional probability shown above. Thus, $P(X, S)$ changes while $P(Y \mid X, S)$ remains invariant. This construction is the setting in which density-ratio importance weighting is theoretically matched. In the hospital analogy, it represents a group-specific change in the distribution of a measured clinical feature without a change in the outcome mechanism conditional on that feature and group.

A.4 Group-specific label noise shift

Finally, the function `label_shift(severity)` modifies recorded labels asymmetrically across groups. Here, `severity` (chosen in $[0, 0.3]$) controls the probability with which clean positive labels for group A and, at half that rate, clean negative labels for group B are flipped. Concretely, we call

- `generate_data(flip_noise_a=severity)`,

which produces $\tilde{Y}$ by flipping positive Y^* labels in group A with probability β and negative Y^* labels in group B with probability $\beta/2$. Features are generated from Y^* before these flips, so the marginal X distribution is unchanged even though the relationship between recorded labels and features is altered asymmetrically. The joint danger-zone experiment applies the α-dependent base rates first and the β-dependent recording noise second, retaining both Y^* and $\tilde{Y}$ for evaluation.

This corresponds to asymmetric diagnostic coding errors across groups in the target hospital (e.g., differential mis-documentation of follow-up need in electronic health records for one group).

Together, these three mechanisms allow us to separately study changes in group-specific base rates, changes in group-specific feature distributions, and changes in group-specific label noise, and to observe how standard fairness and calibration metrics respond when each dimension is varied in isolation.

B Full Fairness and Calibration Metric Definitions

This appendix provides the formal definitions for the metrics summarized in Section 3.2.

We focus on three standard group fairness quantities implemented via `fairlearn.metrics` (`src/metrics.py`):

- **Demographic parity difference (DP diff).** We measure the absolute difference in positive prediction rates between the two groups:

  $$|\Pr(\hat{Y} = 1 \mid S = 0) - \Pr(\hat{Y} = 1 \mid S = 1)|.$$

  A value of 0 corresponds to perfect demographic parity.

  We compute this quantity using `fairlearn.metrics.demographic_parity_difference`. In clinical settings, a large DP difference means that one group is receiving more referrals than the other.

- **Equalized odds difference (EO diff).** We measure the worst-case discrepancy across groups in both true positive and false positive rates:

  $$\max\left(|\mathrm{TPR}_0 - \mathrm{TPR}_1|, |\mathrm{FPR}_0 - \mathrm{FPR}_1|\right).$$

  A value of 0 corresponds to perfect equalized odds.

  We compute this quantity using `fairlearn.metrics.equalized_odds_difference`. In our setting specifically, large EO difference means that either missed follow-up (TPR gap) or unnecessary referrals (FPR gap) are distributed unevenly between groups.

- **True-positive-rate gap (TPR gap).** We separately report $|\mathrm{TPR}_0 - \mathrm{TPR}_1|$, the equal-opportunity violation. This is necessary for interpreting the threshold-adjustment experiment: a smaller TPR gap can coincide with a larger EO difference if the false-positive-rate gap grows. In the one-dimensional label-noise sweep this metric uses recorded outcomes $\tilde{Y}$ and therefore measures disparity relative to the recorded proxy, not latent clinical truth. The joint experiment additionally reports the same quantity using clean synthetic outcomes Y^*.

In addition to group fairness, we track probability calibration at both the global and group levels. We compute the expected calibration error (ECE) following a standard binned estimator. Given predicted probabilities p_i and labels y_i, we partition the interval $[0, 1]$ into B bins, and for each bin b compute the average predicted confidence and empirical accuracy. The ECE is then

$$\mathrm{ECE} = \sum_{b=1}^{B} \frac{n_b}{N} \left| \mathrm{acc}(b) - \mathrm{conf}(b) \right|,$$

where n_b is the number of points in bin b, $\mathrm{acc}(b)$ is the average label, and $\mathrm{conf}(b)$ is the average predicted probability in that bin.

To capture group-wise calibration, we compute ECE separately for each group and report the ECE gap, defined as the difference between the worst- and best-calibrated group:

$$\text{ECE Gap} = \max_a \mathrm{ECE}_a - \min_a \mathrm{ECE}_a.$$

This is implemented in `src/metrics.py` as `calculate_group_ece_metrics`, which returns both the per-group ECEs and their gap. A large ECE gap indicates that for some groups, predicted follow-up risks are systematically over- or under-confident, which may interact with thresholding policies in opaque ways. We also report balanced accuracy to expose utility changes that may accompany an apparent fairness improvement.

C Full Mitigation Strategy Details

This appendix provides the complete implementation details for the three interventions summarized in Section 3.3.

To evaluate whether fairness distortion can be detected or corrected, we implement three post-hoc and training-time interventions:

- **KDE Importance Reweighting.** We estimate the density ratio $w(x) = \hat{p}_{\text{target}}(x)/\hat{p}_{\text{train}}(x)$ using an independent unlabeled target adaptation sample and kernel density estimation (`src/reweighting.py`). The features are jointly standardized, weights are clipped to $[0.1, 10]$, and the logistic-regression loss is reweighted. This approach is most directly justified under covariate shift.
- **Post-hoc Threshold Adjustment (Equal Opportunity).** Using an independent labeled target calibration set $(X, \tilde{Y}, S)$, we take the larger group TPR at threshold 0.5 as the target and choose a deterministic threshold for each group on a grid with spacing 0.01 (`src/adjust_threshold.py`). This attempts to raise the lower-TPR group without retraining the classifier. It does not enforce full equalized odds and can increase the FPR gap. This is analogous to a hospital administrator adjusting group-specific referral thresholds to meet an equal-access policy, while monitoring unintended consequences on other metrics.
- **Target-Distribution Retraining.** As a supervised reference, we retrain the same model family on an independent labeled sample $(X, \tilde{Y})$ from the shifted distribution and evaluate it on the shifted test

set. Group labels are used to compute evaluation metrics, not by the ERM objective. This measures what target-distribution ERM can attain relative to a frozen source model; it is not a universal fairness optimum or a guaranteed improvement.

D Full Implementation Details

This appendix provides the complete experimental implementation details complementing Section 4.1.

Code and reproducibility. Source code, experiment configurations, and plotting scripts are available at `https://github.com/njzfjiang/fairness-shift-simulation`.

- Sample size: source training, shifted evaluation, and target-retraining datasets each use $N = 5000$. KDE adaptation and threshold calibration each use independent samples of size 1000.
- Classifier: Logistic regression on $X = (X_1, X_2)$ without S, with `max_iter=1000` and default L2 regularization (`C=1.0`).
- Fairness and calibration metrics: fairness metrics are computed via `fairlearn.metrics`; ECE uses 10 equal-width bins.
- Repetitions: the one-dimensional diagnostic and mitigation sweeps use three coupled seeds (42, 1042, and 2042); figures report their mean and population standard deviation. The primary 11×10 danger-zone grid uses five coupled seeds (42, 1042, 2042, 3042, and 4042); a cell is marked when at least three seeds satisfy the rule.
- Danger-zone thresholds: 0.05 is a project-defined five-percentage-point materiality floor for TPR-gap worsening; 0.02 treats DP as not materially worse and includes apparent improvement; the factor two captures cases where both metrics worsen but TPR disparity grows disproportionately. These are transparent monitoring conventions rather than clinical or statistical standards. Per-seed metrics are retained so alternative thresholds can be evaluated without rerunning the models.
- Logging: means, standard deviations, raw per-seed metric values, KDE diagnostics, fitted thresholds, clean/recorded TPR gaps, and per-cell danger probabilities are saved in the schema-v5 JSON log.

E Supplementary Results Figures

This appendix provides supporting visualizations for the results summarized in Section 5.1.

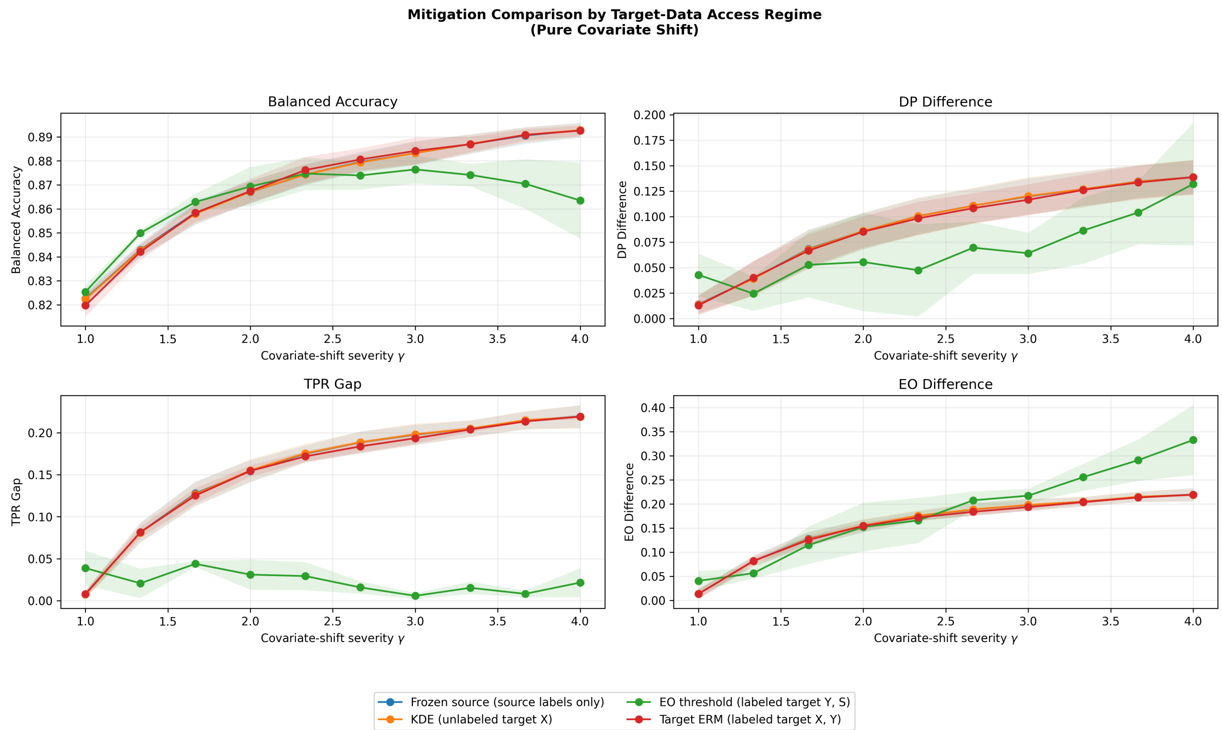

Figure 2: Mitigation comparison under pure covariate shift, organized by target-data access. Lines are three-seed means and bands show population standard deviation. KDE uses unlabeled target covariates; equal-opportunity thresholding uses recorded target outcomes and group labels; target ERM uses labeled target training data. The methods are access regimes rather than interchangeable fixes.

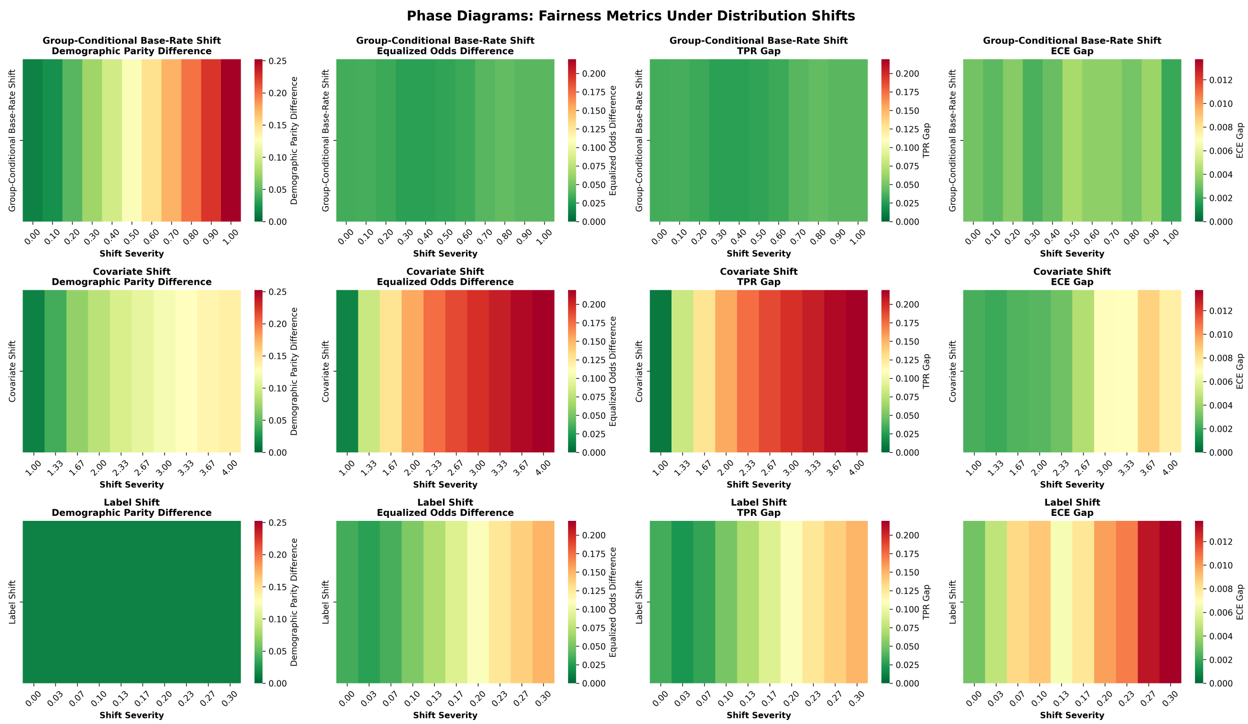

Figure 3: Three-seed mean metrics versus severity for the one-dimensional extensions. Base-rate shift raises DP while the clean-outcome TPR gap changes modestly. Pure covariate shift raises DP and error-based gaps while ECE gap remains small. Under asymmetric label noise, DP remains flat while the TPR gap computed from recorded labels rises.

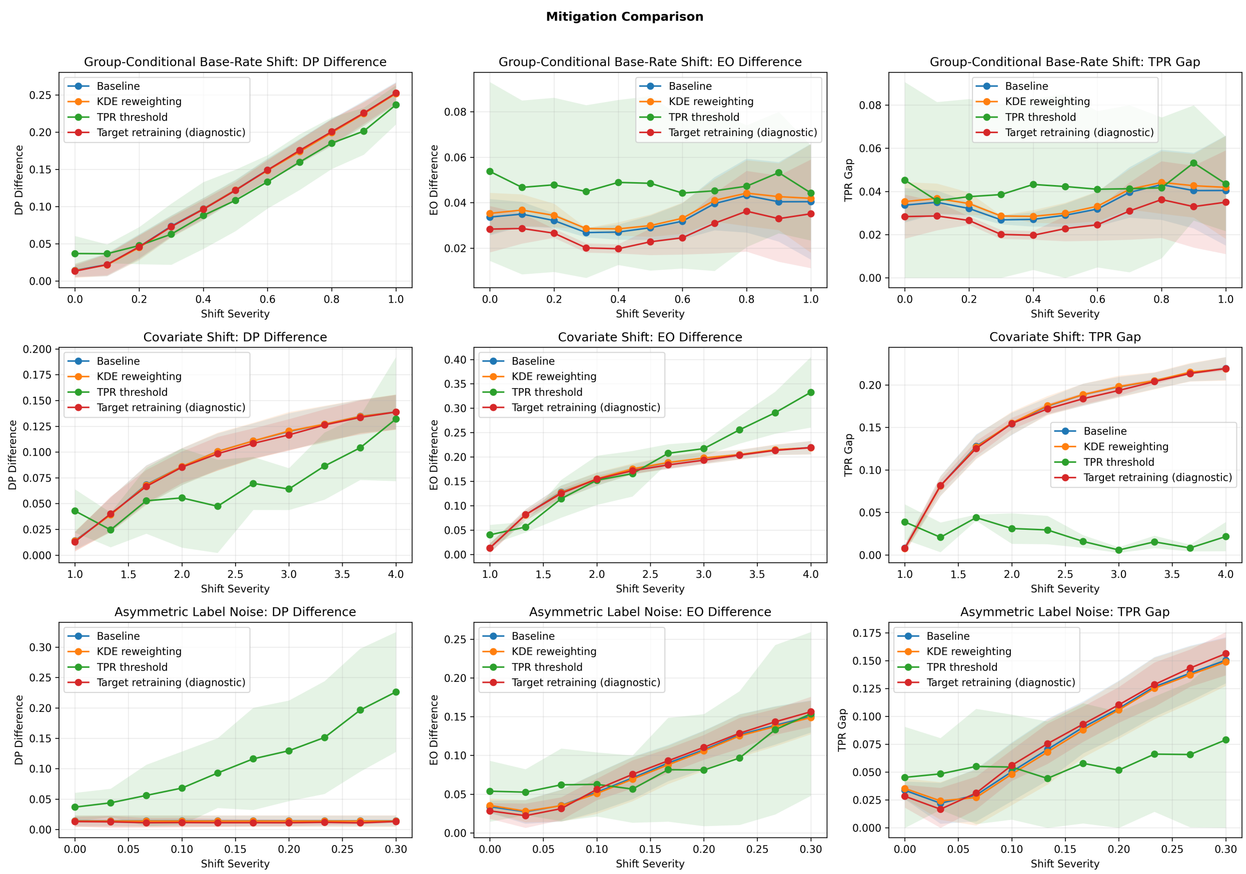

Figure 4: Supplementary three-seed comparison across all shifts; bands show population standard deviation. KDE outside pure covariate shift is a negative control. Threshold adjustment often reduces the TPR gap while increasing DP, FPR-driven EO, or both. Target retraining is a supervised reference, not a fairness-specific mitigation.